

Interpolation Binary Search

Rumus Utama:

$$\text{pos} = \text{low} + ((\text{target} - \text{data}[\text{low}]) / (\text{data}[\text{High}] - \text{data}[\text{low}])) * (\text{High} - \text{low})$$

Kumpulan data : [5, 12, 19, 28, 31, 37, 45, 52, 60, 74, 83, 95]

* Cari Nilai 5

$$\begin{aligned} \text{pos} &= 0 + ((5 - 5) / (95 - 5)) * (11 - 0) \\ &= 0 + (0 / 90) * (11) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Nilai 5 ditemukan Indeks ke-0

* Cari Nilai 12

Iterasi 1

$$\begin{aligned} \text{pos} &= 0 + ((12 - 5) / (95 - 5)) * (11 - 0) \\ &= 0 + ((7 / 90) * (11)) \\ &= 0 + (0.0778 * 11) \\ &= 0 + 0 = 0 \end{aligned}$$

data [0] = 5 < target → geser low ke 1

NO

Date

Iterasi 2

$$\begin{aligned} \text{pos} &= 1 + ((12 - 12) / (95 - 12)) * (11 - 1) \\ &= 1 + (0 / 83) * 10 \\ &= 1 + 0 = 1 \end{aligned}$$

Nilai 12 ditemukan Indeks ke - 1

mencari nilai 89

Iterasi 1

$$\begin{aligned} \text{pos} &= 0 + ((89 - 5) / (95 - 5)) * (11 - 0) \\ &= 0 + (84 / 90) * 11 \\ &= 0 + (0,9333 * 11) \\ &= 0 + 10 = 10 \end{aligned}$$

Indeks Nilai 89 ditemukan Indeks ke - 10